Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 100611

Subject: Computer Architecture

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[2 x 7=14]**

(a) Which of the following is a key characteristic of RISC (Reduced Instruction Set Computer) architecture?

(i) Large number of addressing modes
(ii) Microprogrammed control unit for executing instructions
(iii) Complex instructions with variable lengths
(iv) Execution of one instruction per clock cycle

(b) What is the primary purpose of the system bus in computer architecture?

(i) To transfer data between the CPU, memory, and I/O devices
(ii) To perform high-speed arithmetic calculations
(iii) To store temporary instructions for execution
(iv) To manage and prioritize interrupts from I/O devices

(c) In a multi-level cache hierarchy, what is the role of L1 Cache?

(i) To store data permanently like a hard disk
(ii) To hold instructions for multi-core processors
(iii) To store the most frequently accessed data closest to the CPU
(iv) To manage virtual memory translation

(d) Which of the following storage mediums uses magnetic platters to store data?

(i) Solid-State Drives (SSD)
(ii) Optical disks
(iii) Magnetic Hard Disks
(iv) USB Flash Drives

(e) Which of the following describes the function of an interrupt controller?

(i) It enables the CPU to manage direct memory access operations
(ii) It prevents conflicts between I/O devices and memory
(iii) It schedules tasks for the CPU based on priority
(iv) It handles multiple interrupt requests and prioritizes them for the CPU

(f) What is a major difference between PCI Express (PCIe) and PCI?

(i) PCIe uses a parallel bus, while PCI uses a serial bus
(ii) PCI allows for greater bandwidth than PCIe
(iii) PCIe uses point-to-point communication, while PCI shares a single bus
(iv) PCIe allows for full-duplex communication, while PCI is half-duplex

(g) In the 80x86 architecture, which unit is responsible for fetching instructions and managing data transfer between the CPU and memory?

(i) Execution Unit (EU)
(ii) Arithmetic Logic Unit (ALU)
(iii) Memory Management Unit (MMU)
(iv) Bus Interface Unit (BIU)

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(h) In real mode addressing of the 80x86 architecture, how is the physical address calculated?
(i) By adding the base address to the index register
(ii) By multiplying the segment address by 16 and adding the offset
(iii) By combining the data segment with the stack pointer
(iv) By adding the base address to the offset address

(i) Which of the following best describes a key feature of Digital Signal Processors (DSPs)?
(i) High power consumption for complex calculations
(ii) Specialized instruction sets optimized for real-time data processing
(iii) Support for floating-point arithmetic only
(iv) General-purpose architecture suitable for any application

(j) In MIPS architecture, what type of instruction is specifically used to transfer data from memory to a register?
(i) I-type instruction
(ii) J-type instruction
(iii) D-type instruction
(iv) R-type instruction

Q.2 (a) What is the control unit in a CPU? Discuss how a control unit operates and the difference between hardwired and microprogrammed control units. [7]
(b) What is virtual memory in computer systems? Discuss how virtual memory is implemented and its advantages. [7]

Q.3 (a) Compare CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) architectures. What are their key differences and advantages? [7]
(b) Explain the 80x86 Architecture. What are the main components of the architecture and their functions? [7]

Q.4 (a) Explain integer multiplication and division in computer systems. Discuss how fixed-point and floating-point arithmetic are handled. [7]
(b) Compare IA-32 and IA-64 architectures. Discuss their key features and differences. [7]

Q.5 (a) Discuss the characteristics and uses of magnetic hard disks and optical disks in computer systems. How do they differ in terms of storage mechanism and performance? [7]
(b) Discuss the addressing modes of the 80x86 architecture. How do they enhance the flexibility and efficiency of instruction execution? [7]

Q.6 (a) Describe the role of the Memory Management Unit (MMU) in a computer system. How does it facilitate virtual memory and memory protection? [7]
(b) Describe Instruction Level Pipelining (ILP) and its benefits. Discuss the challenges associated with implementing ILP. [7]

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Q.7 (a) Explain the different types of system memory in a computer. How do primary memory and secondary memory differ in terms of function and characteristics? [7]
(b) What is dynamic scheduling in pipelined processors? How does it improve instruction-level parallelism (ILP) and what are the methods used for dynamic scheduling? [7]

Q.8 (a) Explain the concept of interrupts in computer systems. Discuss how interrupt controllers manage multiple interrupts from different devices. [7]
(b) Explain VLIW (Very Long Instruction Word) architecture. Discuss its features and advantages in instruction-level parallelism (ILP). [7]

Q.9 (a) What is Direct Memory Access (DMA)? Explain the working of a DMA controller and its role in I/O operations. [7]
(b) Explain System-on-Chip (SoC) architecture. Discuss its benefits and applications in modern computing systems. [7]

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